

Noise Cancellation & Equalizer Headphones

Tony Antony

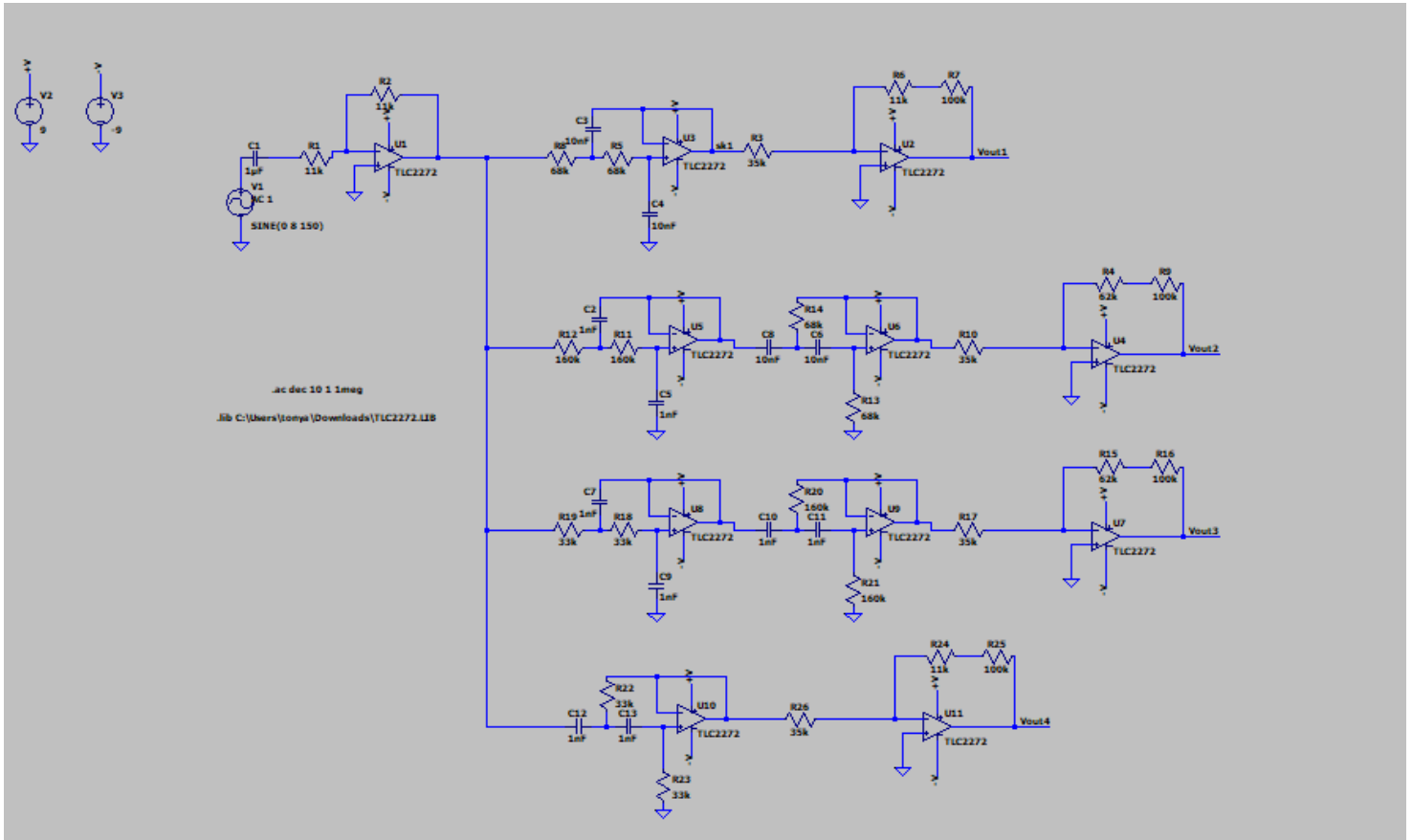
December 14th 2024

Introduction

My project was to design and assemble a printed circuit board that acts as a medium between an input source and the output source which is the headphones. I wanted to be able to achieve two things: noise cancellation and equalizing ability. For the equalizer, it will receive music input from another source and be able to amplify or filter out certain ranges of frequencies. For the noise cancellation, it will use two mics (one for the left headphone output and the other for the right) and shift the signal out of phase effectively cancelling the ambient noise surrounding you. The overall design contains two different circuits, one for the noise cancellation and the other for the equalizer, which will be combined at the end and fed into a pair of headphones through an audio jack.

Circuit Description

Since there are two different stages to my circuit I will break it down into modular parts.



(Image of the Equalizer circuit)

Here is the equalizer circuit which makes use of Sallen-Key filters to create four different frequency ranges to either amplify or attenuate. I chose a Sallen-Key configure because they are 2nd order active filters with -40db/decade slope outside its cutoff range. First the signal is fed through a high pass filter to filter out any DC component and to provide enough signal power for the four stages.

First Stage:

A low pass filter in a Sallen-key configuration.

High frequency cutoff: $1 / (2\pi * 68k * 10nF) \approx 235 \text{ Hz}$

So this stage is focused on the range 20 Hz to 235 Hz (Bass, kick drums, deep melodies, etc)

Second Stage:

A Band Pass filter with two Sallen-key configurations where one is low and the other is a high pass filter.

Low frequency cutoff: $1 / (2\pi * 68k * 10nF) \approx 235 \text{ Hz}$

High frequency cutoff: $1 / (2\pi * 160k * 1nF) \approx 995 \text{ Hz}$

So this stage is focused on the range 235 Hz to 995 Hz (Midrange, vocals, guitar, etc)

Third stage:

A Band Pass filter with two Sallen-key configurations where one is low and the other is a high pass filter.

Low frequency cutoff: $1 / (2\pi * 160k * 1nF) \approx 995 \text{ Hz}$

High frequency cutoff: $1 / (2\pi * 33k * 1nF) \approx 4825 \text{ Hz}$

So this stage is focused on the range 995 Hz to 4825 Hz (Soprano vocals, High pitch piano keys, etc)

Fourth Stage:

A high pass filter in a Sallen-key configuration.

Low frequency cutoff: $1 / (2\pi * 33k * 10nF) \approx 4825 \text{ Hz}$

So this stage is focused on the range 4825 Hz to 20k Hz (High frequency noise, Clarity, etc)

All these stages go through a Gain stage using four simply inverting op amps one for each band.

I used a 100k logarithmic pot for a variable resistance to change the gain and my goal was to swing the gain from -10db to 10db.

Calculate the voltage gain (A_v) for the decibel levels required:

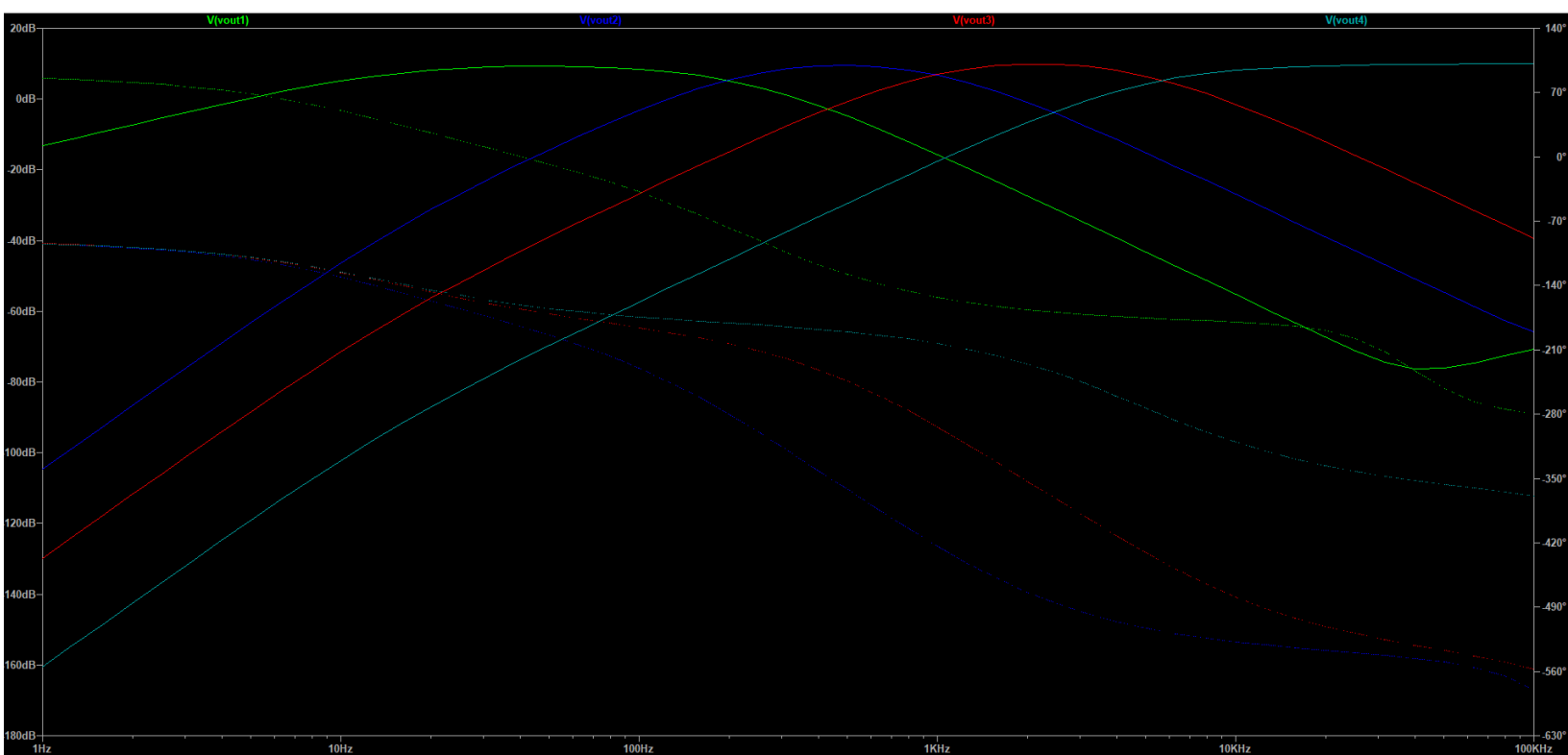
$$20\log_{10}(A_v) = -10\text{db} \quad A_v \approx 0.316$$

$$20\log_{10}(A_v) = 10\text{db} \quad A_v \approx 3.16$$

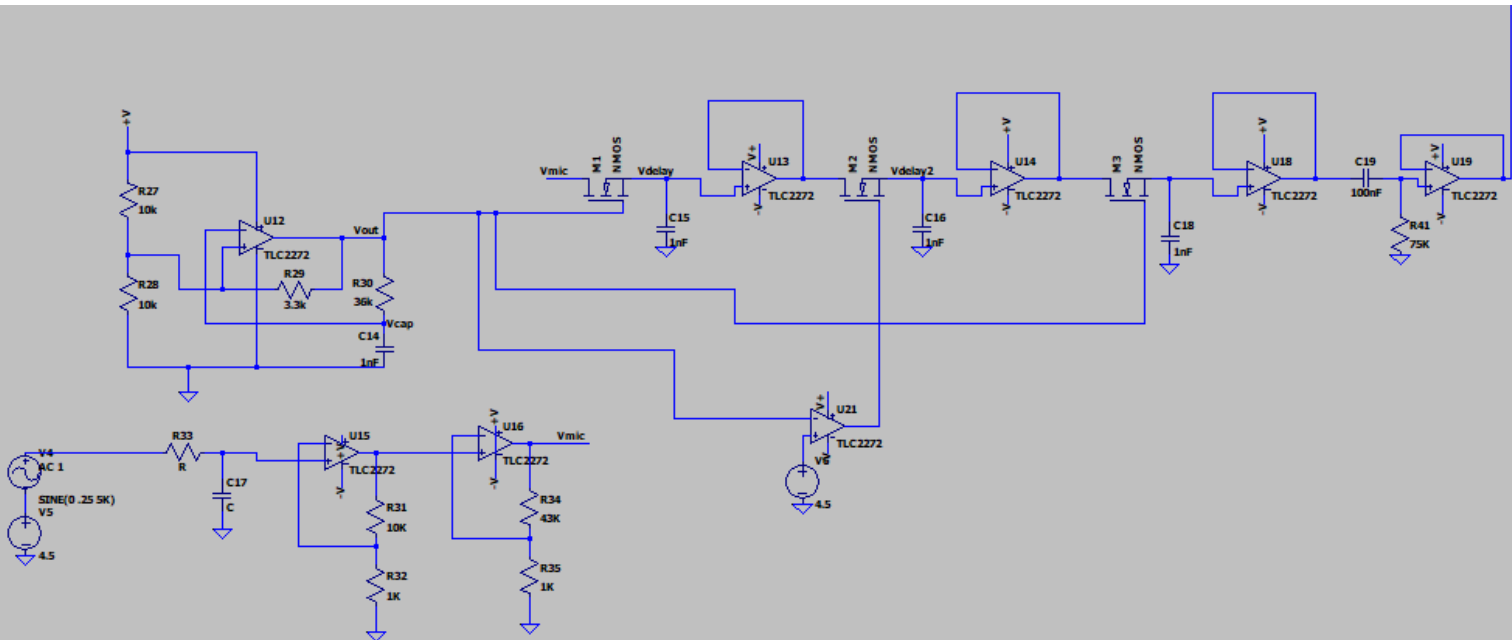
I chose the resistances for the inverting op amp to be 35k and (11k+Pot)

If pot is turned all the way off, the gain will be $11k/35k \approx 0.314$

If pot is turned all the way on, the gain will be $111k/35k \approx 3.17$



This picture shows an AC sweep from 1 to 100k Hz of the four equalizer outputs. You can see the four graphs pertain to their respective frequency range. This picture shows when the pot is turned all the way on so you can see that they all are currently at 10db gain.



Here is the circuit for one of the mics for noise cancellation. The other mic has the exact same circuitry. First I had to figure out a gain for my mic in order to make it usable for my circuit. I used the CMA-4544PF-W electret condenser microphone with sensitivity levels of -44dbV/Pa.

$$V_{out} = 10^{(SPL - 94 + SdBV)/20} V$$

So voltage levels would range from the microVolts to the milliVolts depending on the sound pressure.

I decided to use a total gain of 430 using two non-inverting op amps to make a notable sound for the headphones. The mic is biased at 4.5 volts for the mosfet delay stage. I also carefully selected this gain so that the voltage levels can be within a certain level for the mosfet delay stage which I will talk about later.

Once I had the gain set I can't just yet simply invert the signal (or shift it 90 degrees) in order to create a signal that will interfere destructively with the outside noise. There is one more thing to do before this, which is to delay the signal from the mic due to the distance between the mic and your ears as shown in the picture below.



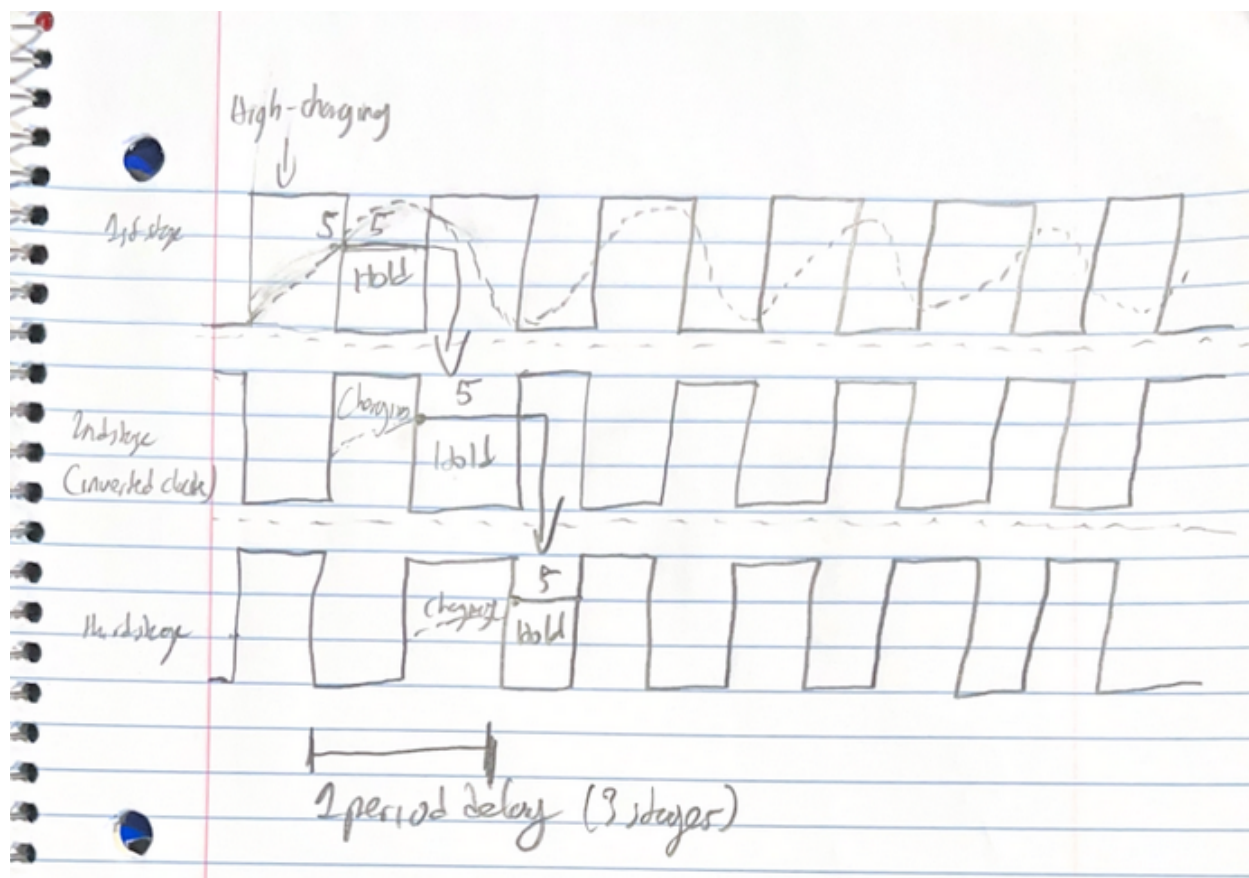
The distance is about 2 to 3 cm as shown above (depends on how close I tape the mic to the headphones).

Speed of sound: $343 \text{ m/s} = 34300 \text{ cm/s}$, Delay time (lowest) = $2/34300 \approx 58.3 \mu\text{s}$

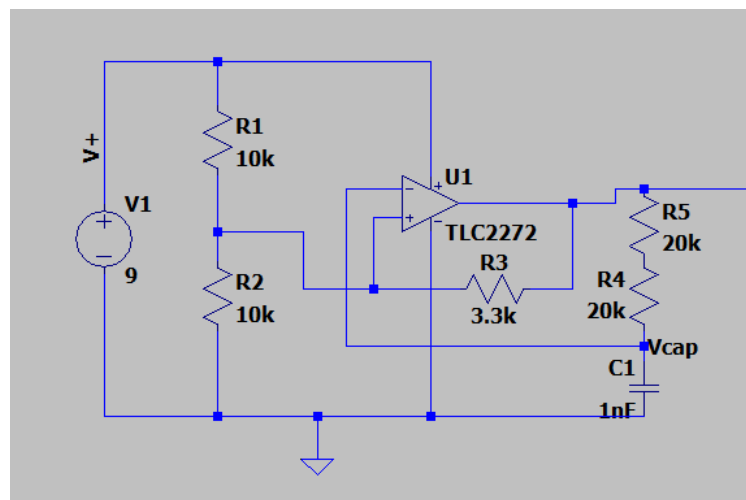
Delay time(highest) = $3/34300 \approx 87.3 \mu\text{s}$

You might be asking yourself is it that big of a deal. “We are only dealing with low frequencies like 20Hz to 20kHz”, is something you might think. However something like a 2500 Hz signal has a period of about 400 μs and so a 87.3 μs delay most definitely will lead to a not ideal outcome.

In order to delay the sound I decided to use a three stage stop and hold circuit using a mosfet, capacitor, and op-amp configuration as seen in the circuit diagram above. To make an analogy think of a latch in digital logic. To create a circuit that propagates a signal that is edge sensitive to a clock signal, you need a flip-flop. Each mosfet-capacitor combination corresponds to a latch and so to create a flip-flop design I need two different clock signals: a square wave clock and its inverted signal. The delay will correspond to the period of the clock signal. Below is a diagram of how my three stage signal will cause a delay of one total period corresponding to the clock signal



Since I was using the TLC2274 op-amp for my equalizer circuit, I realized I can simplify my design to make an oscillator with the same type of op-amp. And so I decided to make a relaxation oscillator.



$$T = -2 R4C1 \ln \frac{R3}{R3 + R}$$

There are two things to consider: The delay must be in the appropriate range and the frequency must be at least twice the highest frequency in the input (Nyquist Criterion)

I decided to use a 20k linear pot with a constant resistance of 20k.

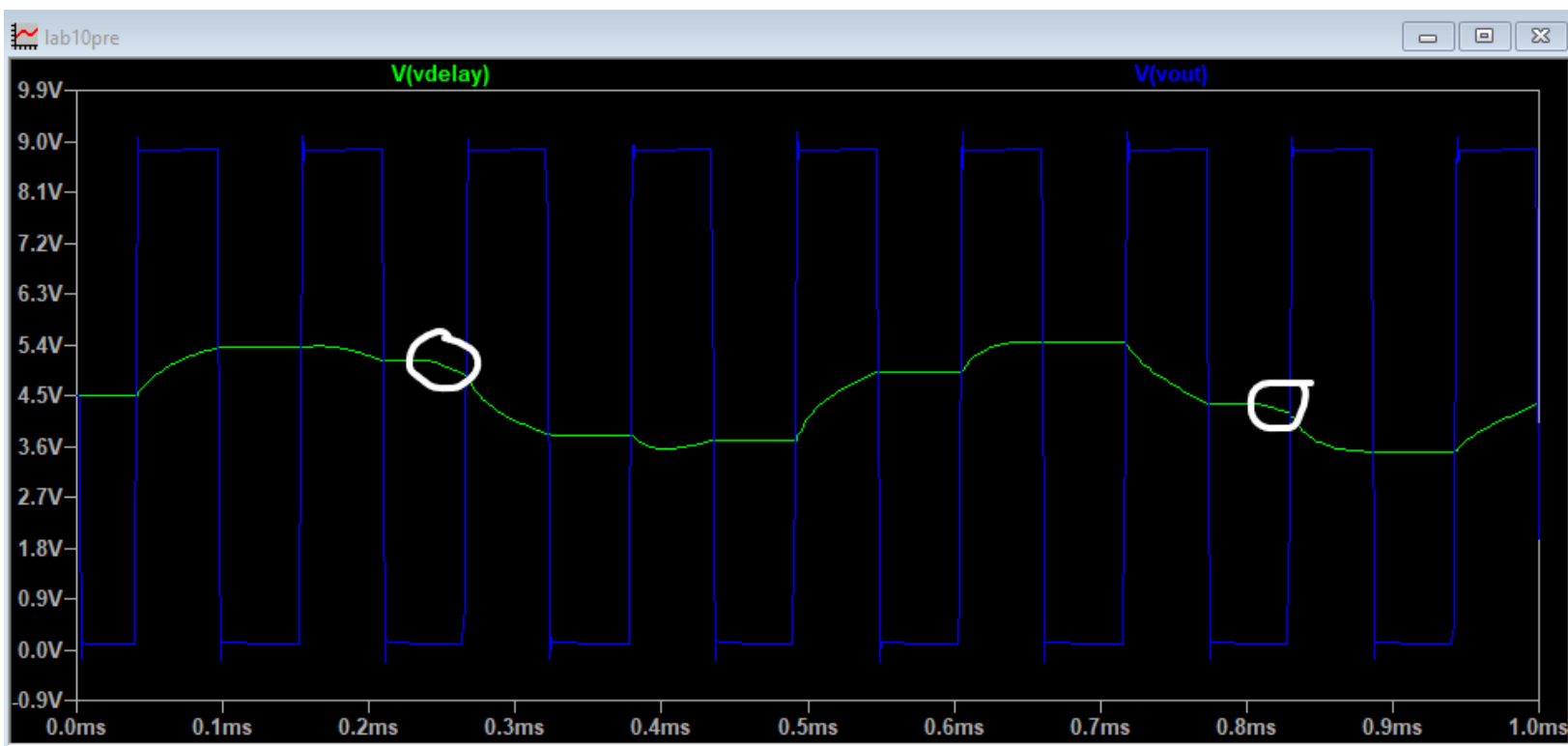
If the pot is turned all the way on: $R_4 = 40k$ | $T(\text{period}) \approx 111.5\mu s$ | $F = 1/T \approx 8968 \text{ Hz}$

If the pot is turned all the way off: $R_4 = 20k$ | $T(\text{period}) \approx 55.75\mu s$ | $F = 1/T \approx 17936 \text{ Hz}$

Since I designed my delay stage to have a delay of 1 period the numbers above are corresponding to the required delay.

I designed the noise cancellation to only work to about 3400 Hz and so the minimum sampling frequency of 8968 Hz corresponds to the Nyquist Criterion.

Now to make it work for the delay stage I must bias the mic signal to center it from +V(9V) to ground since that is where the clock operates and so I used a voltage divider and biased the mic to 4.5 volts. I also need to be careful as to not create enough gain for the mic or else the mosfet can be in reverse biased mode and cause the capacitor to discharge from source to drain even though I want it to hold its voltage when the VG(gate voltage) is at 0(GND). The image below shows this non-ideality.



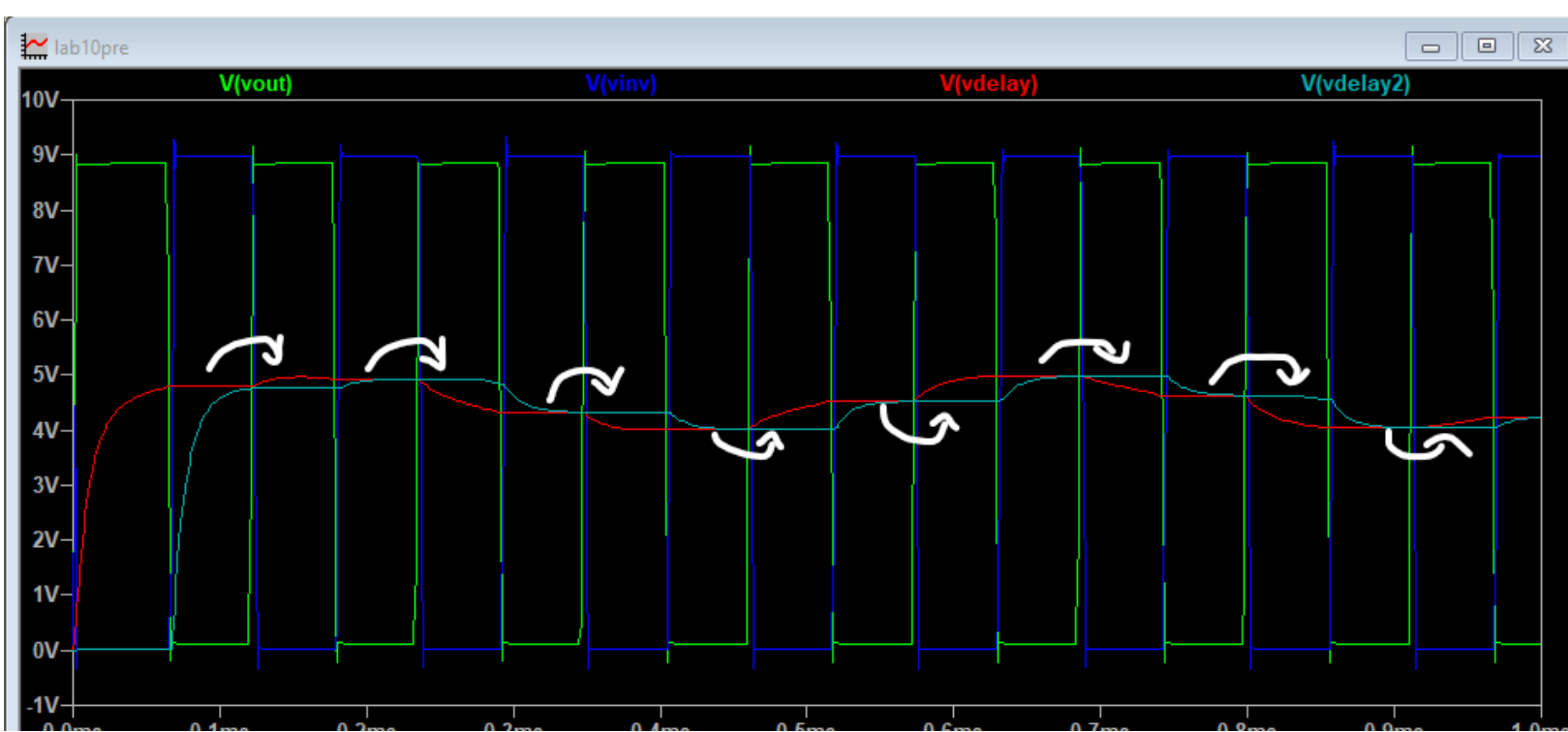
The circled parts shows the capacitor discharging even in its hold time (when the clock is 0) V_{pp} was at 1 volt which is higher than the LTspice VSD reverse biased threshold voltage

To counter this I must keep the mic signal after its gain to within the VSD threshold for reverse biased mode. I used the 2N7002KT1G onsemi transistor with the VSD threshold at around 0.8 to 1.2 volts.

DRAIN-SOURCE DIODE CHARACTERISTICS

Forward Diode Voltage	V_{SD}	$V_{GS} = 0\text{ V},$ $I_S = 200\text{ mA}$	$T_J = 25^\circ\text{C}$	0.8	1.2	V
			$T_J = 85^\circ\text{C}$	0.7		

The gain I used of 430 keeps the voltage to within these thresholds of a 0.8 to 1.2 V peak to peak(V_{pp}) and so it should resolve this problem.



Here is a snapshot of the delay with two stages. As seen in my handwritten diagram the signal shifts half a clock cycle with 2 delay stages. Since the simulation is extremely slow with three we can assume with the third stage it will delay an entire clock cycle. I also changed V_{pp} to 0.5.

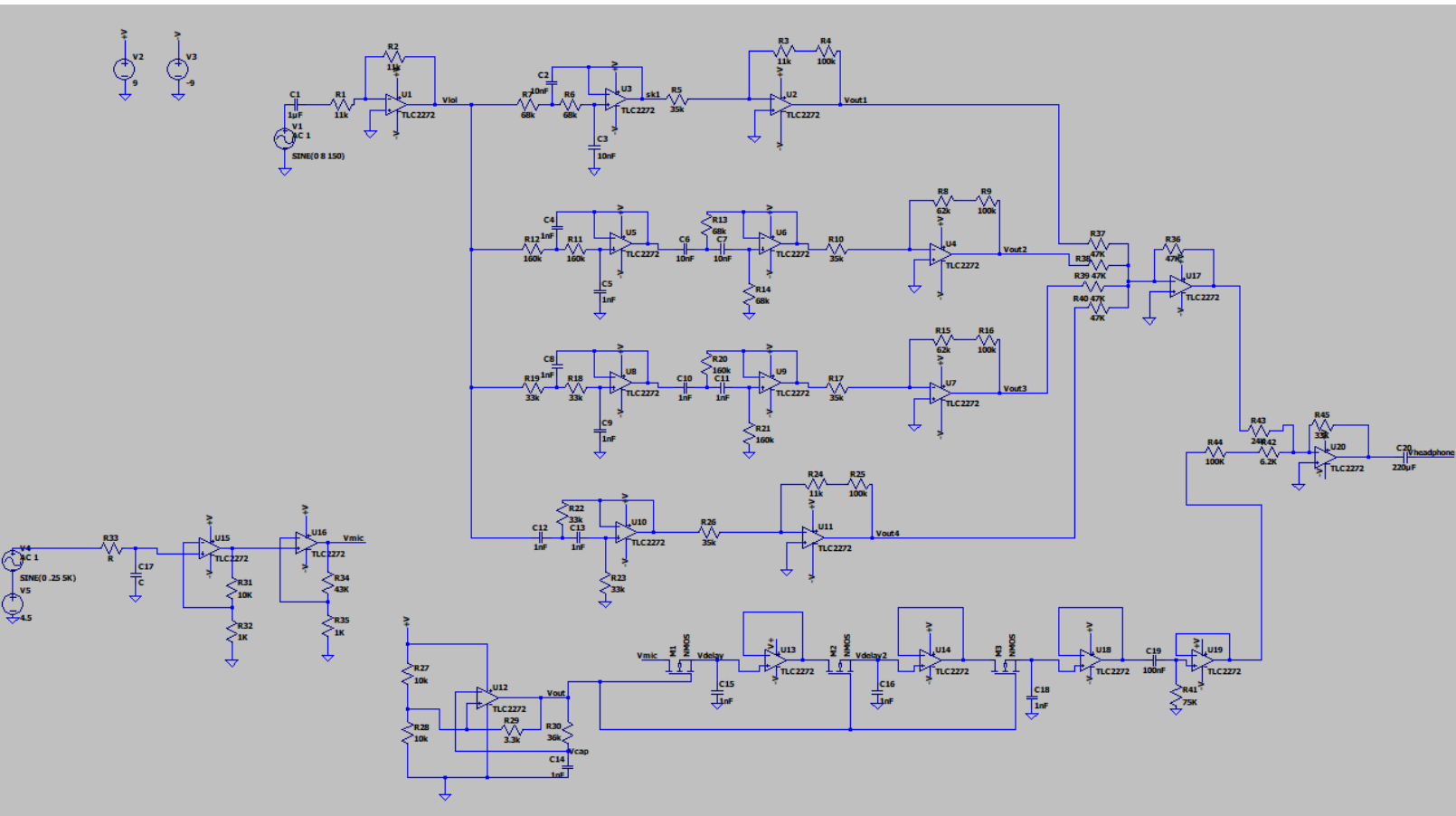


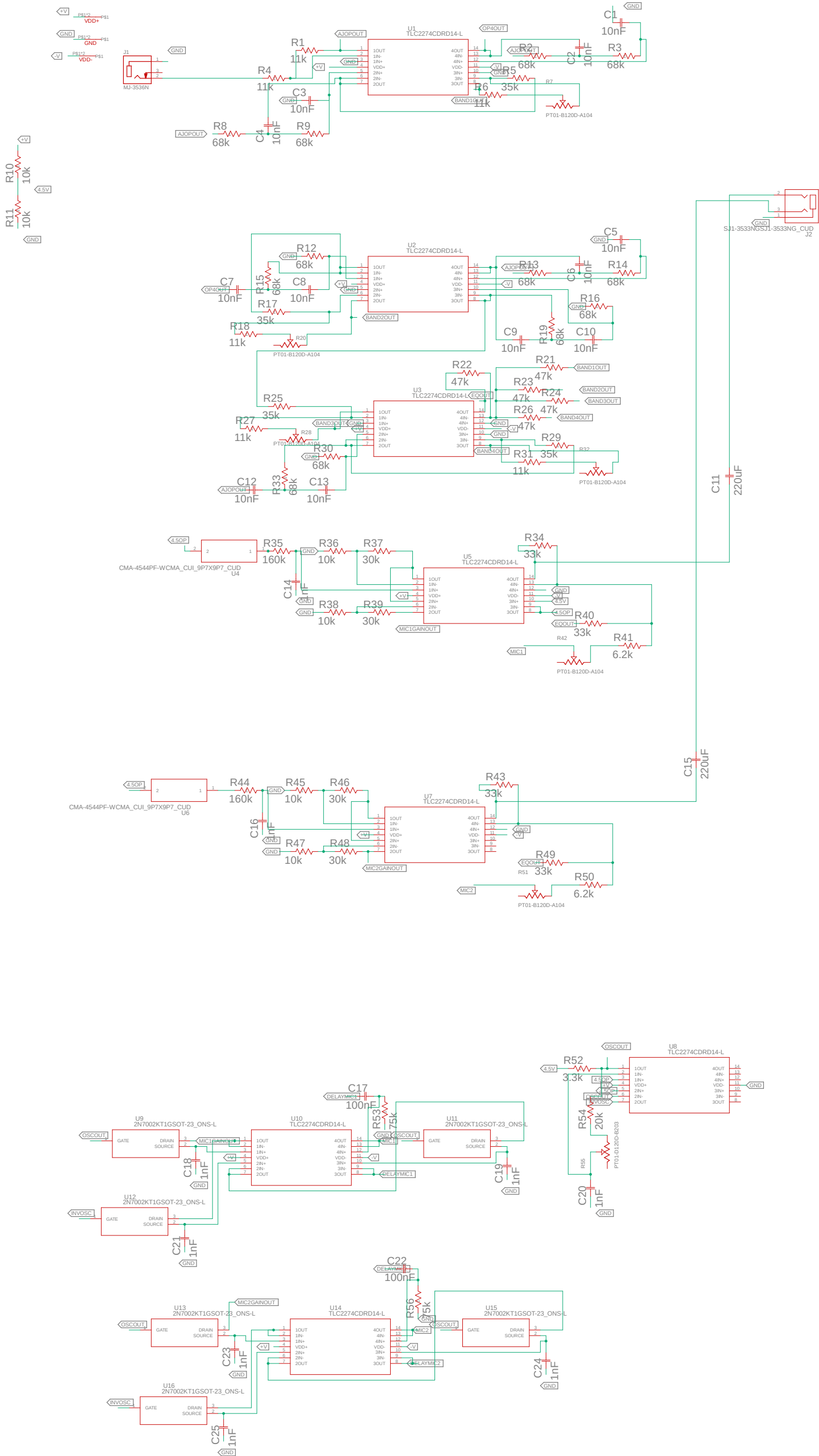
Image of the entire circuit with the summing amplifier

Now you might be asking where is the mic signal inverted or shifted out of phase???

In the final stage where the signals are combined I used an inverting amplifier with a 100k log pot variable gain which will invert the signal and create noise cancellation.

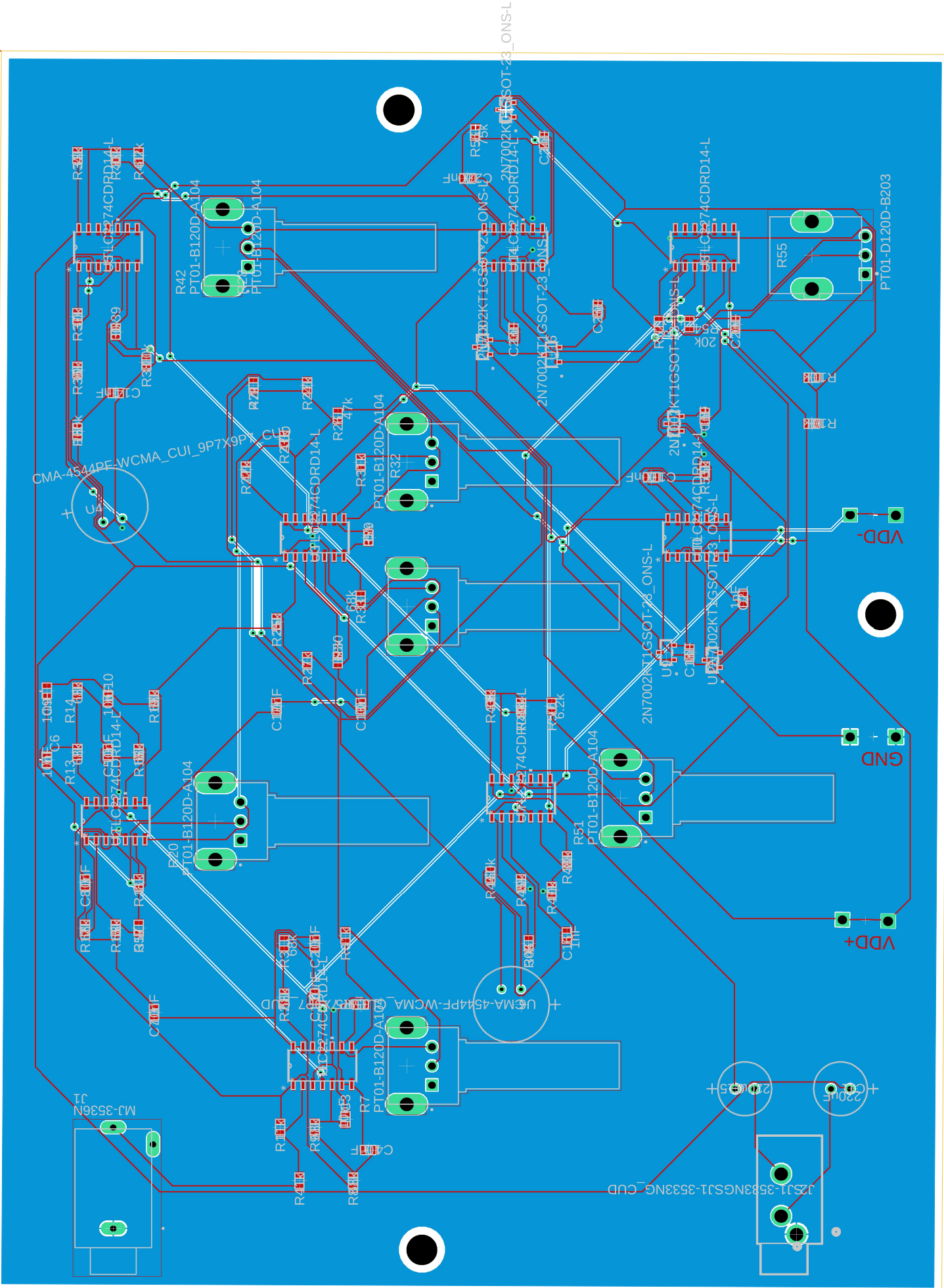
Bill of Materials

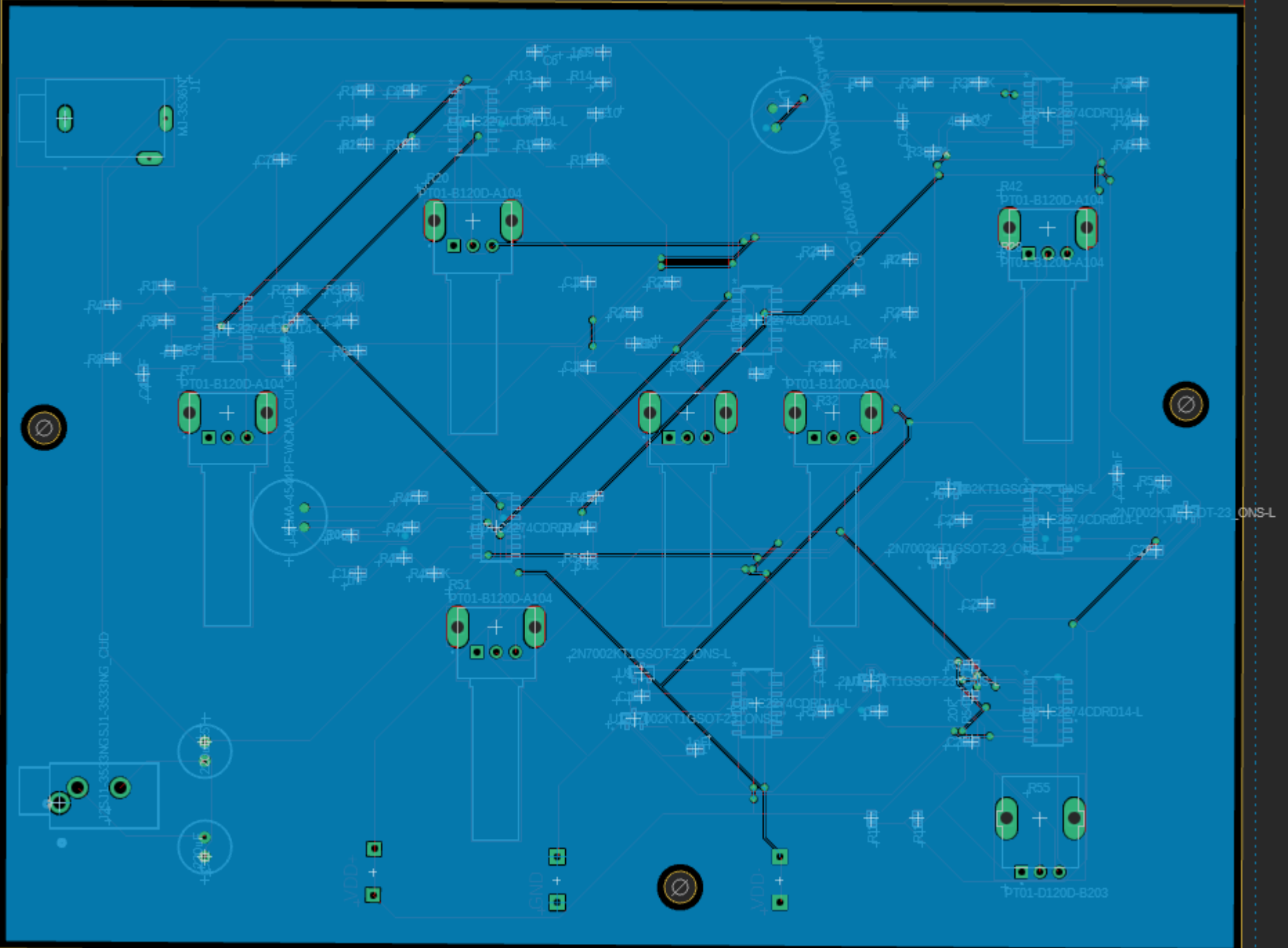
Description	Manufacturer	Manufacturer part number	Vendor	Vendor part number	quantity	Cost
Microphone	Same Sky	CMEJ-0415-42-LP	Digikey	102-6541-ND	2	1.34
Op Amp	Texas instruments	TLC227x	Digikey	296-1814-5-ND	8 chips of 4 op amps	17.28
0805 Capacitor	Provided	Provided	Provided	Provided	42	Provided
0805 Resistor	Provided	Provided	Provided	Provided	16	Provided
Mono Audio jack	Same Sky (Formerly CUI Devices)	MJ-3536N	Digikey	CP-3536N-ND	1	0.78
Aux cord	MORELECS	N/A	Amazon	N/A	2	Already Had
Stereo Audio jack	Kycon, Inc.	STX-3000	Digikey	2092-STX-3000-ND	1	1.25
Mosfet transistor for sample and hold	Onsemi	2N7002KT1G	Digikey	2N7002KT1GOSTR-ND	6	1.14
Logarithmic Variable Resistor	Same Sky (Formerly CUI Devices)	PT01-B120D-A104	Digikey	2223-PT01-B120D-A104-ND	6	6.78
Linear Variable Resistor	Same Sky (Formerly CUI Devices)	PT01-D120D-B203	Digikey	2223-PT01-D120D-B203-ND	1	1.13
Electrolytic Capacitor	Rubycon	25YXJ220M6.3X11	Digikey	1189-2913-ND	2	0.58



162.23

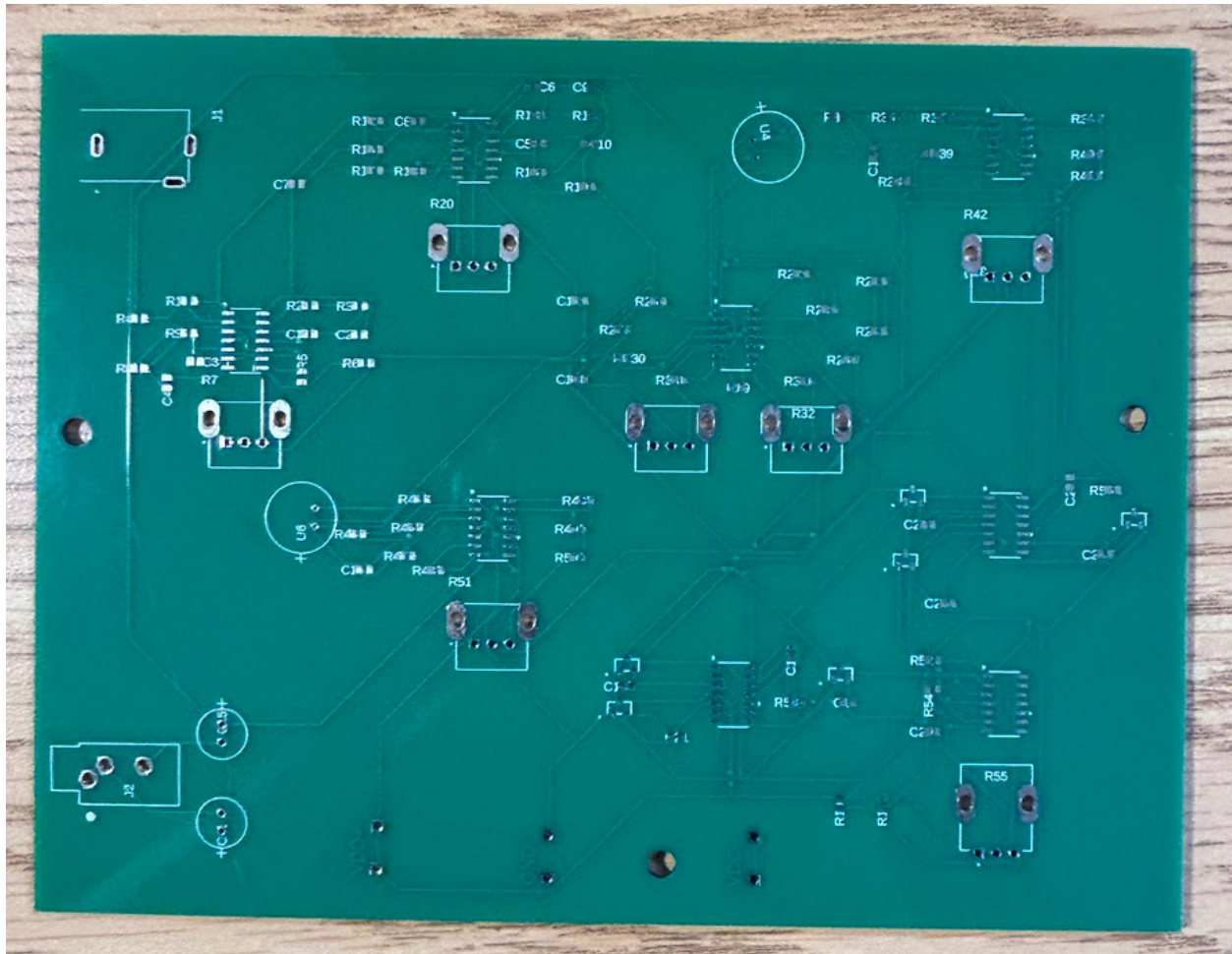
124.01



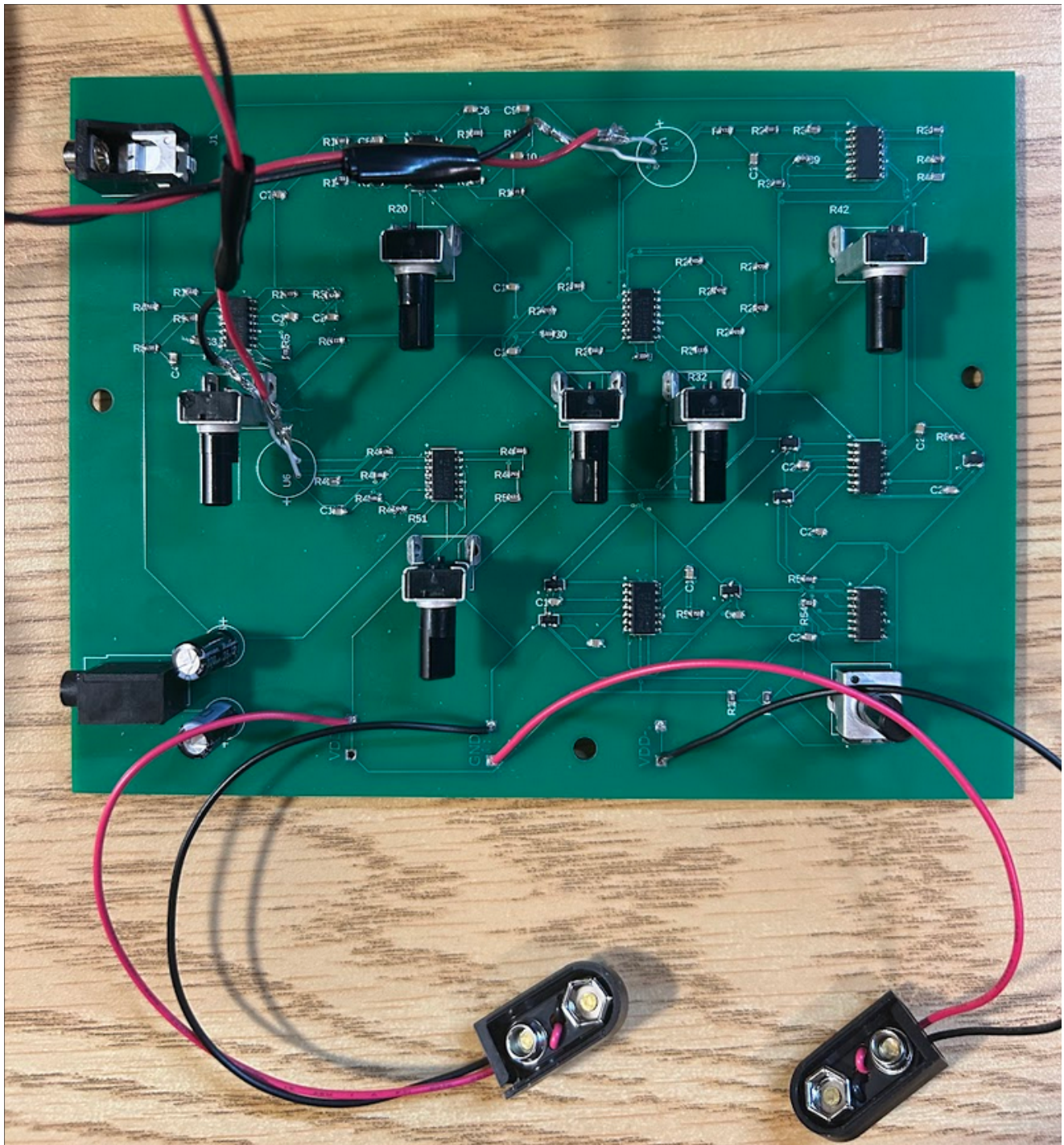


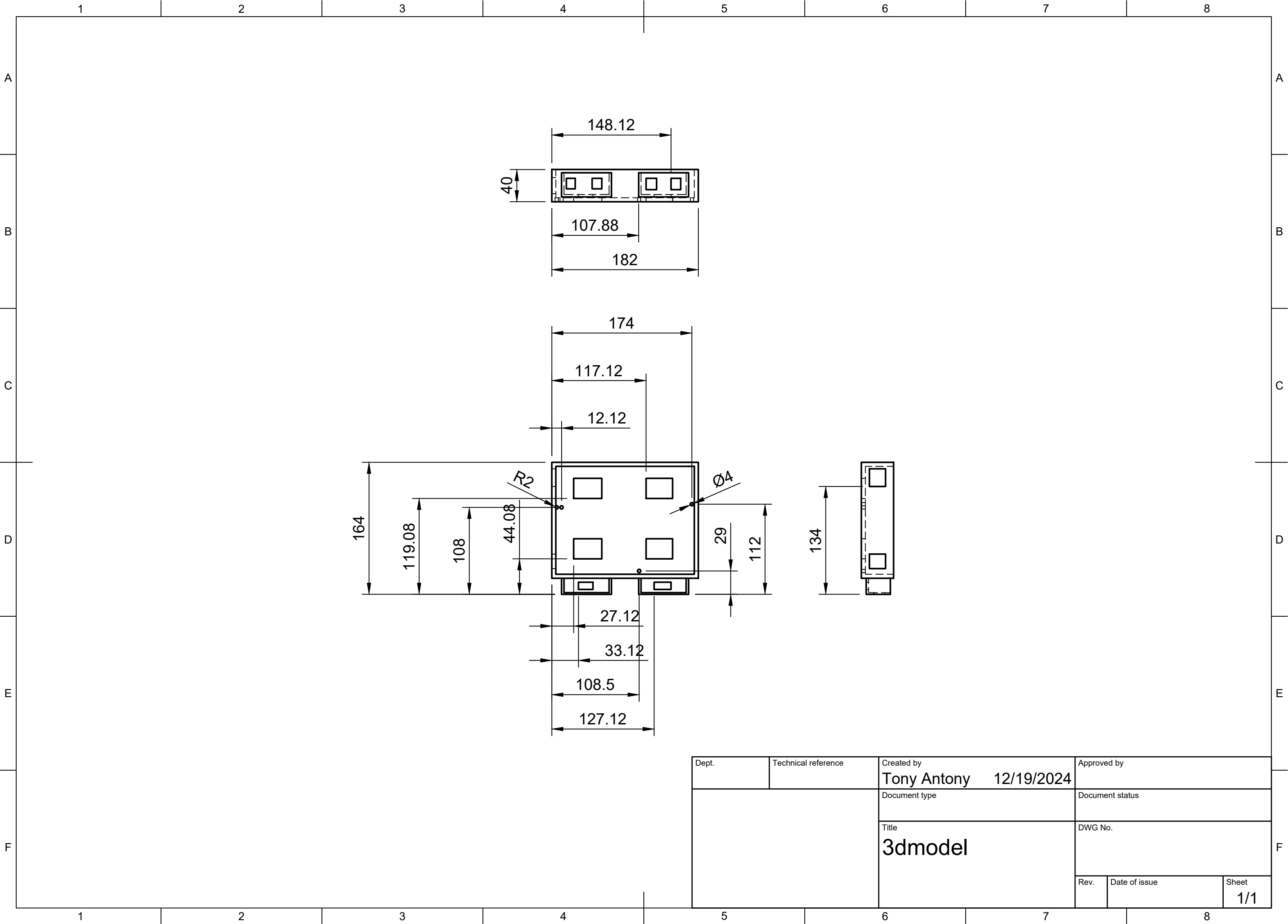
The above shows the Schematic Diagram and Board layouts

Unpopulated board

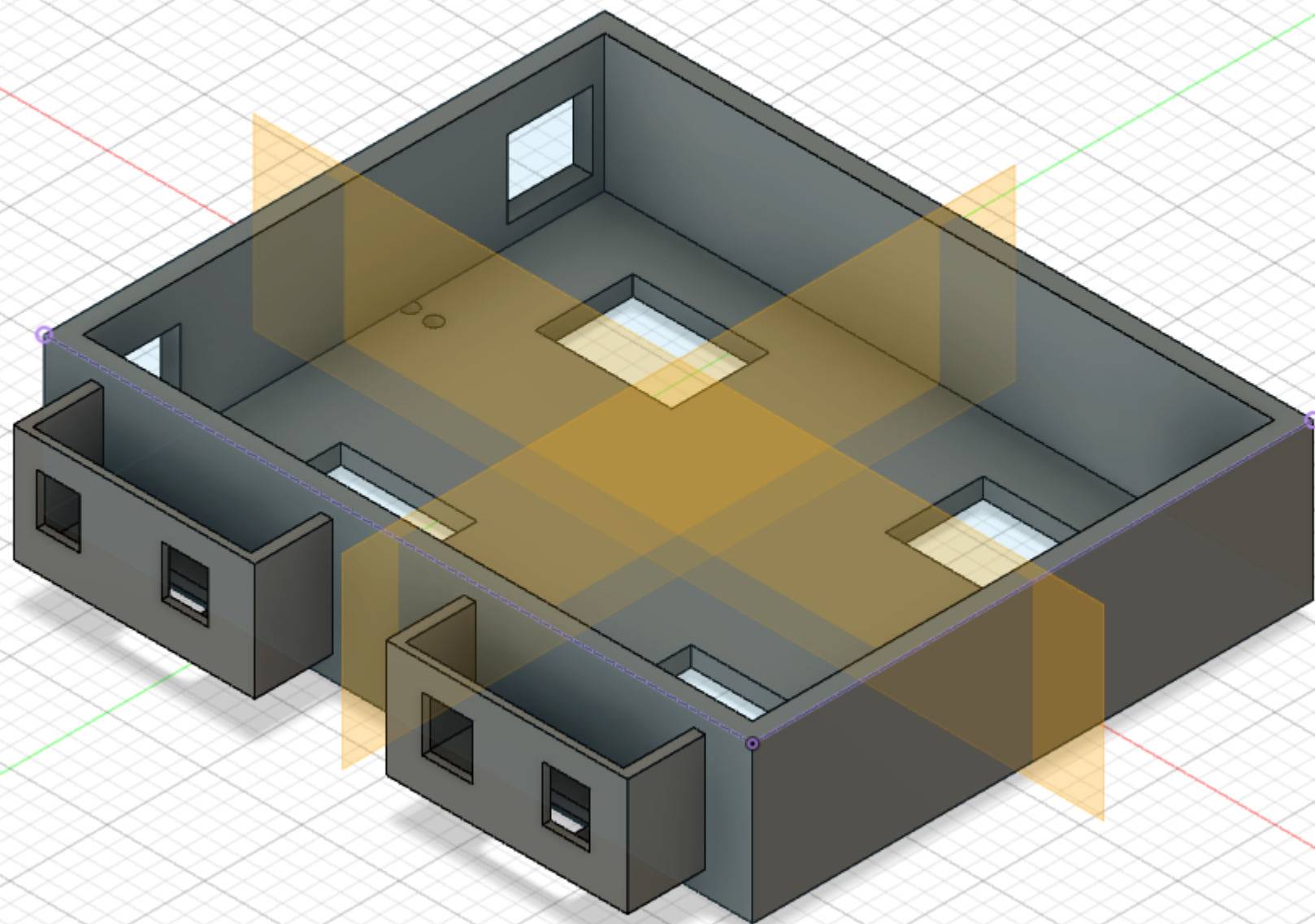


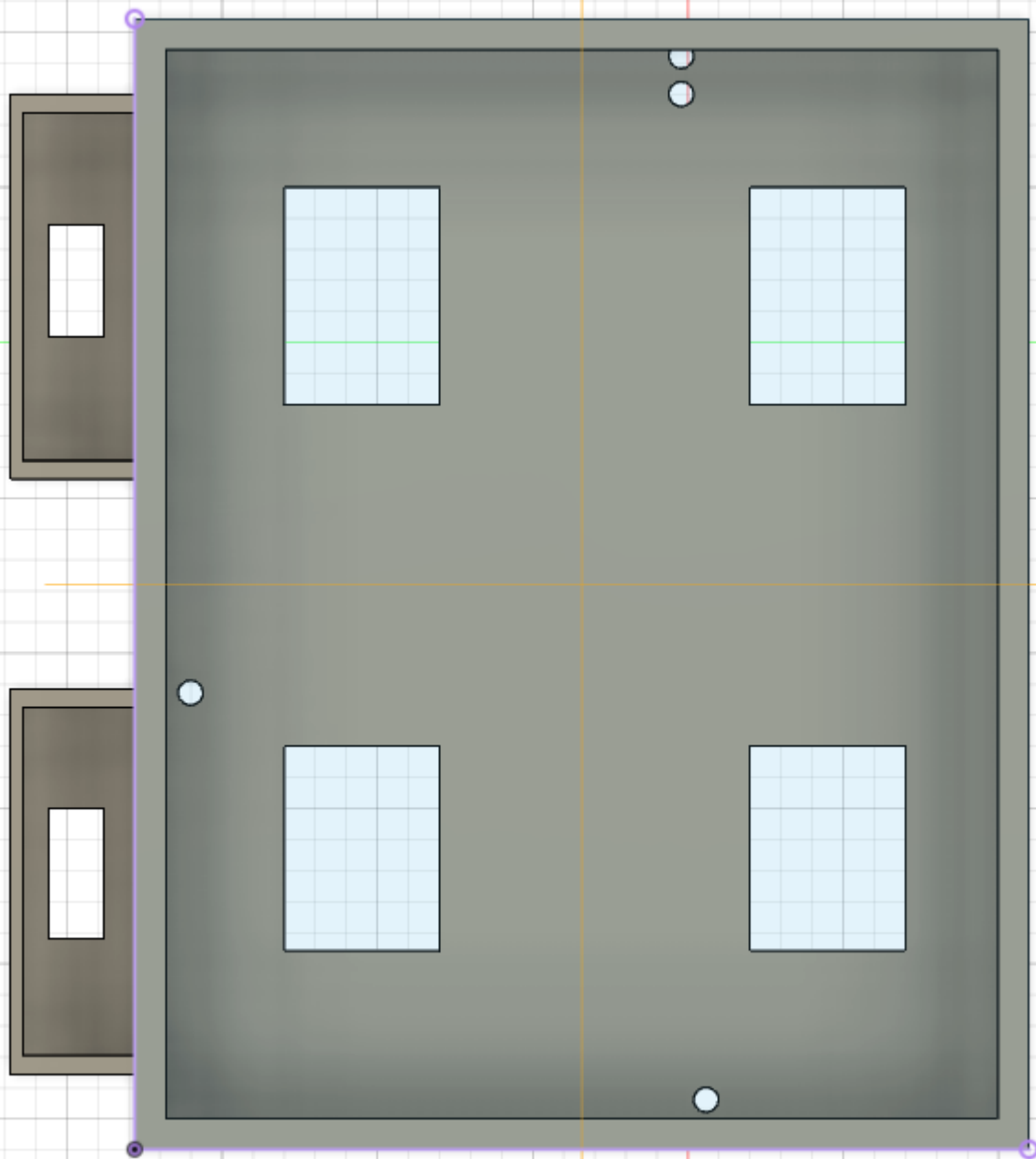
Built board

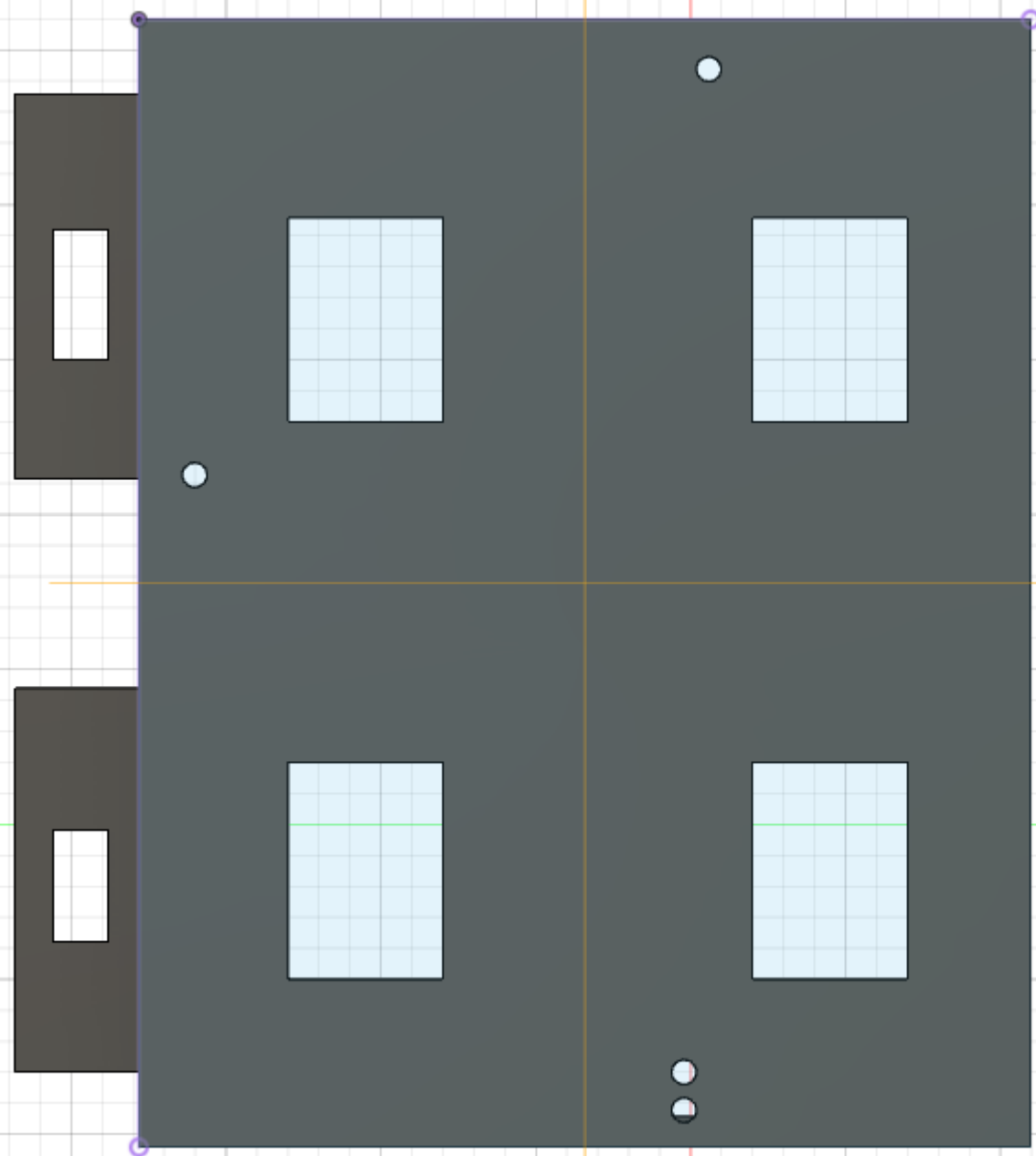


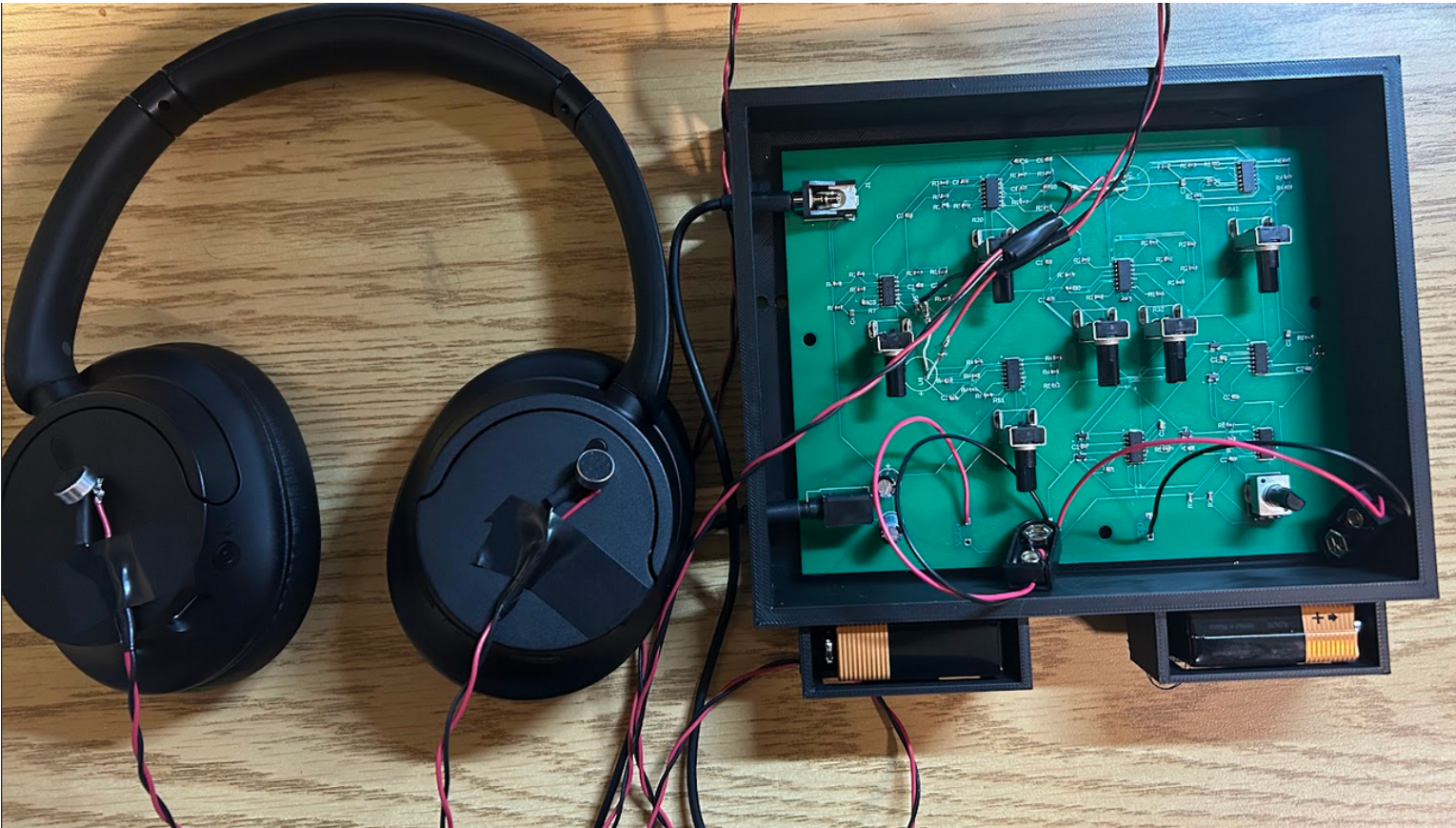


Dept.	Technical reference	Created by Tony Antony 12/19/2024	Approved by	
		Document type	Document status	
		Title 3dmodel	DWG No.	
			Rev.	Date of issue
			Sheet 1/1	



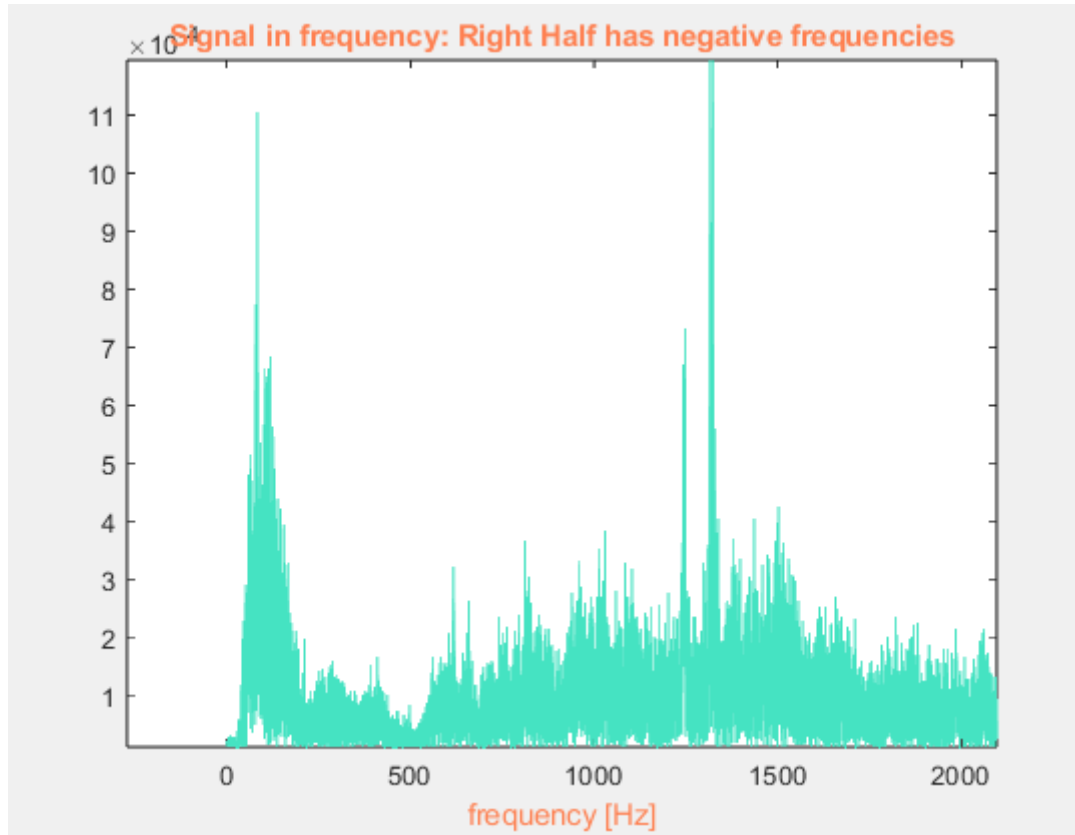




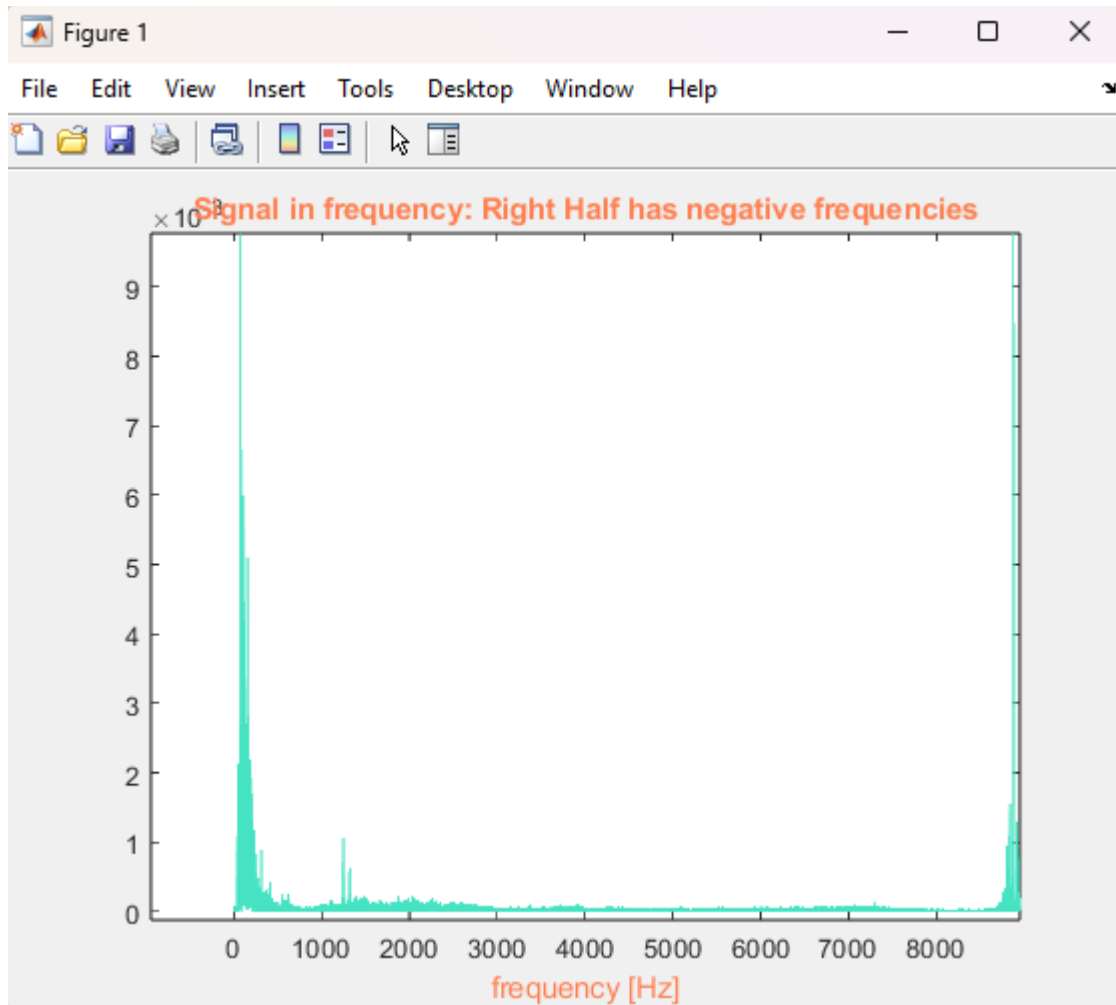


Results

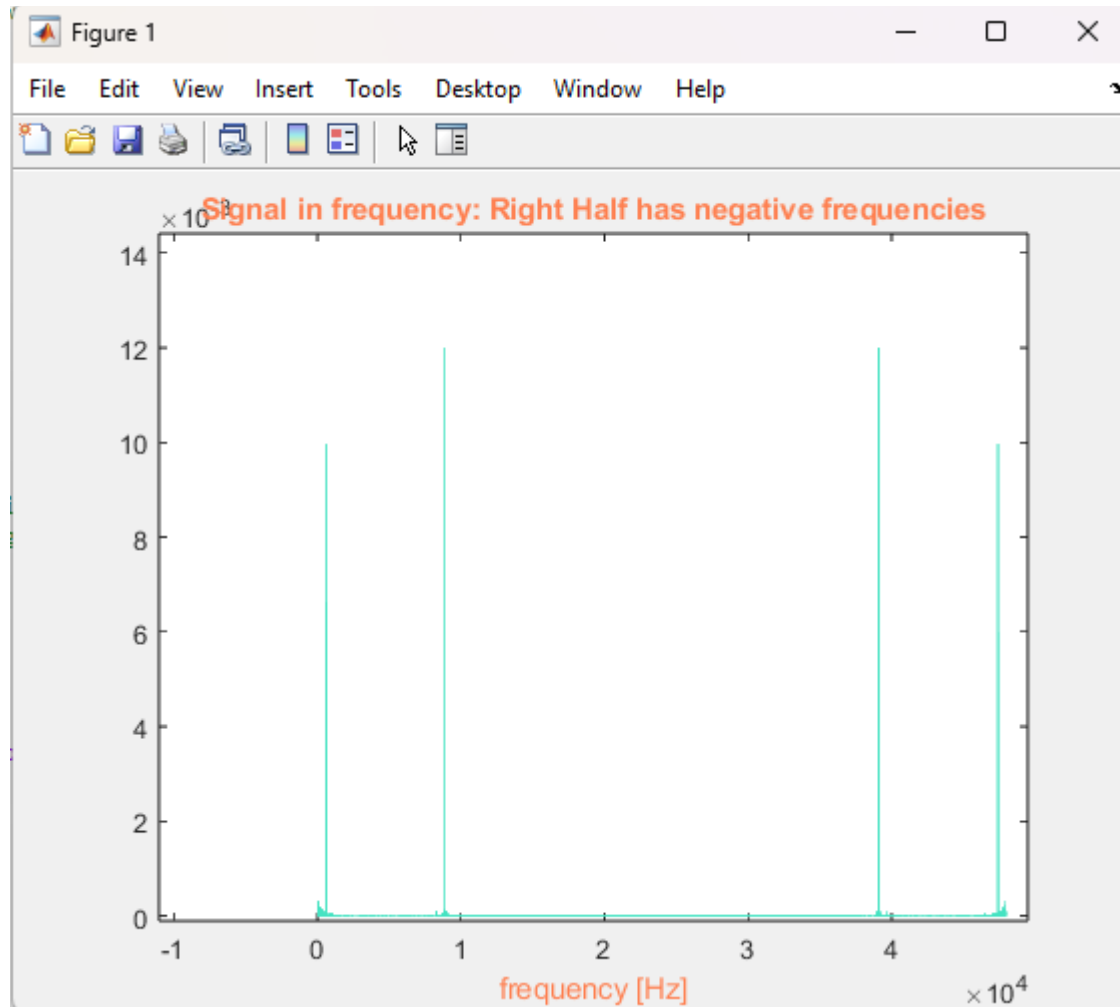
Equalizer worked as expected and each frequency band was filtered or gained based on pot value.



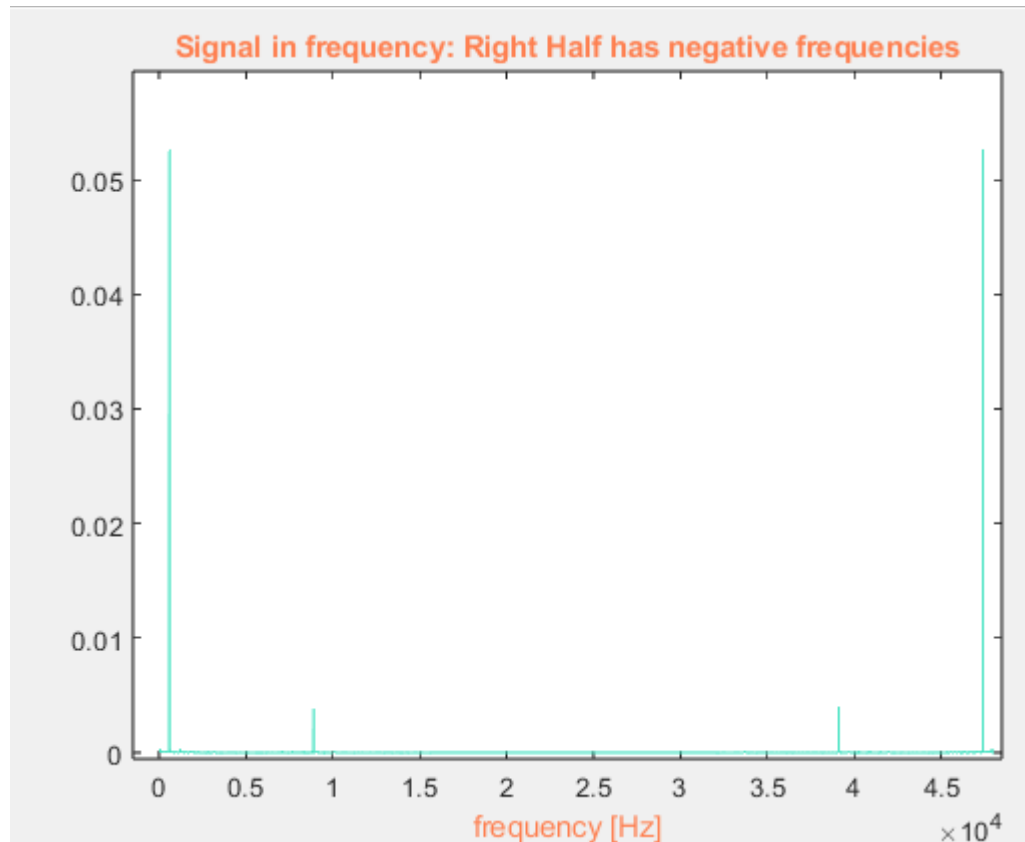
This is the fourier analysis output with Band1 (20Hz to 250 Hz) in attenuation mode.



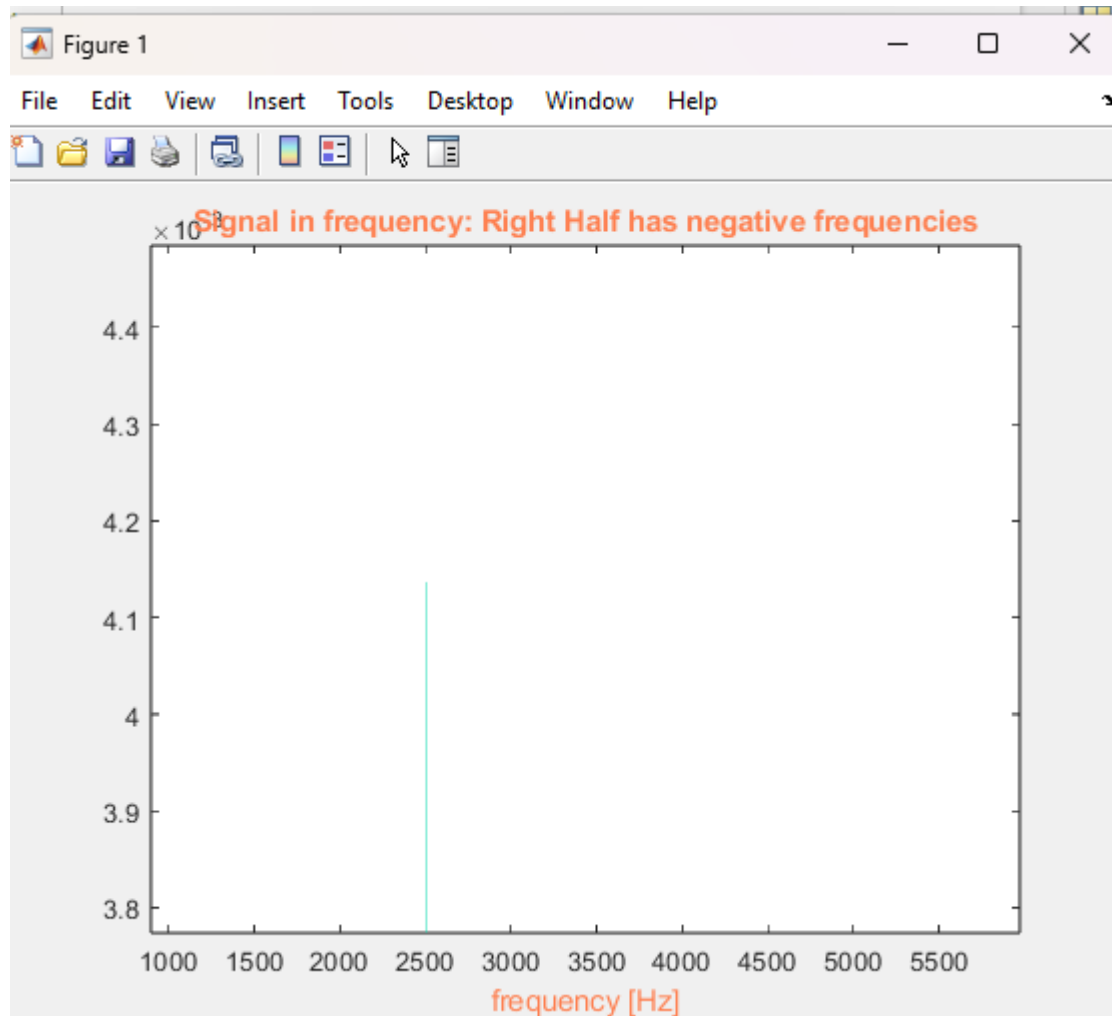
This is with Band1 in full gain mode. As you can see the amplitudes for 25Hz to 250Hz have significantly increased.



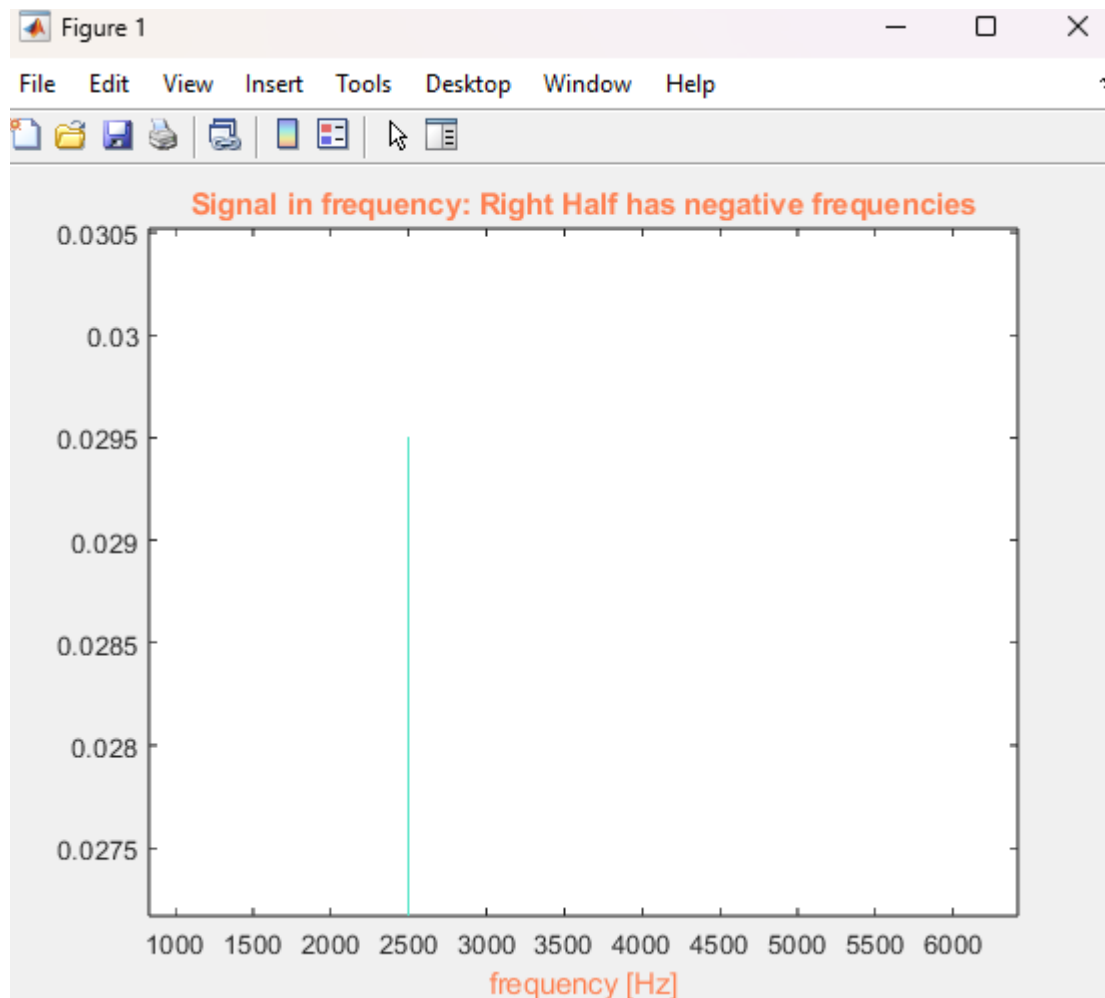
Band2 in attenuation mode (Signal is a pure 600 Hz signal and amplitude is the furthest to the left for the 600Hz.)



Band2 in full gain mode, as you can see the gain is significant. The amplitude furthest to the left has went from 10×10^{-3} before, to .05 right here.



Band3(1000Hz to 5000Hz) in full attenuation mode. Signal was a pure 2500 Hz wave.
Amplitude: 4.15×10^{-3}



Band3 in full gain mode. As you can see the amplitude increased from 4.15×10^{-3} to 0.0295 as seen here.

My Band 4 which is from 5k to 20k hertz also worked well to cut off brightness, clarity and airiness of the sound.


```

% Fourier Transform (fft) of an audio file
%*****
format long

[y,Fs] = audioread('audio.wav'); %read audio file with sam

sound(y,Fs);
|
nsize = length(y);
sampling_interval = 1/Fs;
TStop = nsize/Fs;
ta = 0:sampling_interval:TStop-sampling_interval;

subplot(2,1,1);
plot(ta, y, 'Color', [.27 .89 .76]);
xlabel('time [sec]', 'Color', [1 .5 .314]);
title('Signal in time', 'Color', [1 .5 .314]);

f0 = 1/TStop; %fundamental frequency fft will show harmonics of f0
Sf = (1/nsize)*fft(y); % this line does the fft; (1/nsize) is the scaling
deltaf = f0; %deltaf is similar to the sampling interval for our fft graph
FStop = nsize*deltaf; %total sample time for fft graph
fa = 0:deltaf:FStop-deltaf;

subplot(2,1,2);
plot(fa, abs(Sf), 'Color', [.27 .89 .76]);
xlabel('frequency [Hz]', 'Color', [1 .5 .314]);
title('Signal in frequency: Right Half has negative frequencies', 'Color', [1 .5 .314]);

```

I ran this code above for the FFT analysis

Noise Cancellation Results

Noise Cancellation did not work. :(

Using a scope I saw that after the mic gain stage somehow the output was clipped at the rails (9V). I will spend more time debugging the gain stage.

The scope showed the oscillators and delay stage worked as expected although it just used corrupted mic signals at 9 volts which was eventually filtered out using a DC block at the end.

Ending Statements

What I would do differently is have a greater magnitude of attenuation. I only went to -10db and it didn't fully cancel out the corresponding frequency band.

I would also buy a mic with leads as it was a pain wiring them and I don't know if they work properly with all the heat from soldering.

I would also make a variable gain for the mic gain stage since it was after the gain stage of the mic where the signal was corrupted for noise cancellation. I am still debugging it at this moment so I am not fully sure what the true error is.

I would buy a potentiometer pointing up so I can make a proper enclosure for the pcb case. The pots were all parallel to the board so I can't make an enclosure for the case without removing it all the time if I want to move a pot.

These are all the changes on the top of my head. I am sure there's more!

Class advice: I honestly liked this class alot! It ran well for the subject topic and how the focus is on a project that pretty much tests everything about modern circuit board prototyping. I believe that a PCB class should be little theoretical and more application based (where we can apply theory from other classes) and this class was just that.